Chapter 10: One- and Two-Sample Tests of Hypotheses (Part III)

ENGR 0021

Categorical Data Tests (Ch. 10.11–10.14)

Goal

- Analyze categorical (count) data
- Compare observed vs expected frequencies

Tests Covered

- Goodness-of-Fit
- Independence
- Homogeneity

Key Tool

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Chi-Square Goodness-of-Fit Test

Purpose

- Does data follow a specified distribution?

Hypotheses

$$H_0: \text{ data follow specified distribution}$$

$$H_1: \text{ data do not follow distribution}$$

Test Statistic

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Degrees of Freedom

$$df = k - 1 - (\text{parameters estimated})$$

Expected Frequencies

Expected Count

$$E_i = np_i$$

Requirements

- All $E_i \geq 5$ (approximate rule)
- Independent observations

Loaded Die Example

A gambler is accused of using a loaded die, but he pleads innocent. A record has been kept of the last 60 throws. There is disagreement about how to interpret the data and a statistician is called in.

Value	Observed Frequency	Expected Frequency
1	4	10
2	6	10
3	17	10
4	16	10
5	8	10
6	9	10
Sum	60	60

Solution

Chi-Square Test vs z-test

Chi-Square (χ^2) Test

- Use when **the full distribution matters**
- Tests whether data match a population with **specified proportions**
- Die Example:
  - Are outcomes 1–6 equally likely (each 1/6)?

z-Test

- Use when **only the average matters**
- Tests whether data come from a box with a given **mean**
- Die Example:
  - Is the average of the box 3.5?

Summary

- χ^2-test: checks **entire distribution**
- z-test: checks **mean only**

TV Show Statistics Discussion

In The Big Bang Theory, Sheldon claims he used a chi-square test to determine that 32, 17, and 5 are most likely.

(a) Does this make sense?

TV Show Statistics Discussion

(b) What is the chi-square test used for?

Extension: What if the Wheel is Biased?

Discussion Question:
What if the roulette wheel is actually biased?

- Suppose some numbers occur more often than others
- Would a chi-square test detect this?

Chi-Square Test for Independence

Purpose

- Are two categorical variables related?

Hypotheses

$$H_0: \text{ variables are independent}$$

$$H_1: \text{ variables are dependent}$$

Expected Counts

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{n}$$

Degrees of Freedom

$$df = (r-1)(c-1)$$

Handedness Example: Observed Counts

	Men	Women	Total
Right-handed	934	1070	2004
Left-handed	113	92	205
Ambidextrous	20	8	28
Total	1067	1170	2237

Test whether handedness and gender are independent.

$$H_0 : \text{Handedness and gender are independent}$$

$$H_1 : \text{Handedness and gender are not independent}$$

Handedness Example: Expected Counts

Expected counts are computed using

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$$

For example,

$$E_{\text{Right, Men}} = \frac{(2004)(1067)}{2237} \approx 956.2$$

	Men	Women
Right-handed	956.2	1047.8
Left-handed	97.8	107.2
Ambidextrous	13.4	14.6

Handedness Example: Chi-Squared Statistic

The chi-squared test statistic is

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2 = \frac{(934-956.2)^2}{956.2} + \frac{(1070-1047.8)^2}{1047.8} + \frac{(113-97.8)^2}{97.8} + \frac{(92-107.2)^2}{107.2} + \frac{(20-13.4)^2}{13.4} + \frac{(8-14.6)^2}{14.6}$$

$$\chi^2 \approx 11.9$$

Degrees of freedom:

$$(df) = (r-1)(c-1) = (3-1)(2-1) = 2$$

Handedness Example: Conclusion

Chi-Square Test for Homogeneity

Purpose

- Do multiple populations have the same distribution?

Hypotheses

$$H_0: \text{ all populations have same distribution}$$

$$H_1: \text{ at least one differs}$$

Key Idea

- Same calculation as independence test
- Different interpretation

Independence vs Homogeneity

Same Math, Different Questions

Test	Question
Independence	Are variables related?
Homogeneity	Are populations similar?

Case Study: Two-Sample Categorical Data

Setup

- Compare categorical outcomes across groups
- Use contingency table

Steps

1. State H_0 (same distribution / independence)
2. Compute expected counts
3. Compute χ^2
4. Compare to critical value or p-value

Summary

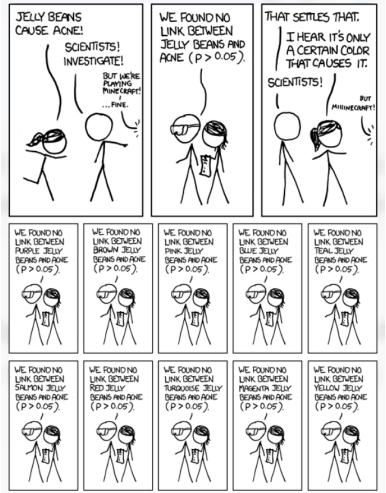
Three Chi-Square Tests

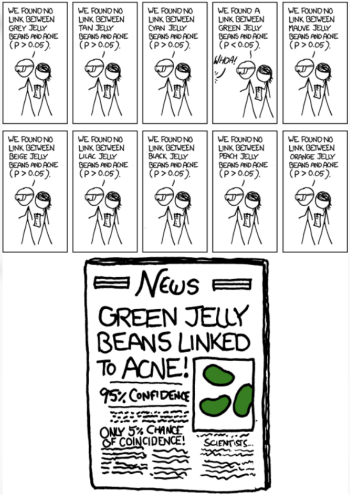
- Goodness-of-fit $\rightarrow$ one distribution
- Independence $\rightarrow$ relationship between variables
- Homogeneity $\rightarrow$ compare populations

Key Formula

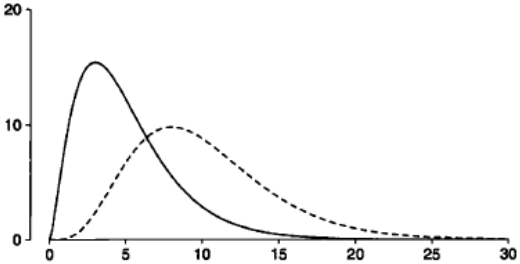
$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Comic Statistics Discussion





Chi-Squared



(a) One of these is $df = 5$ and one is $df = 10$. Which is which?
(b) If $\chi^2 = 15$, the P-value will be bigger with _____ degrees of freedom than with _____ degrees of freedom. Options: 5, 10
(c) If there are 10 degrees of freedom, the P-value will be bigger with $\chi^2 =$ _____ than with $\chi^2 =$ _____. Options: 15, 20